(18) $\int e^{-4x}\cos x\,dx$ is

(1) $\frac{e^{-4x}}{17}[4\cos x-\sin x]+c$ (2) $\frac{e^{-4x}}{17}[-4\cos x+\sin x]+c$

(3) $\frac{e^{-4x}}{17}[4\cos x+\sin x]+c$ (4) $\frac{e^{-4x}}{17}[-4\cos x-\sin x]+c$

(19) $\int \frac{\sec^2 x}{\tan^2 x-1}dx$

(1) $2\log\left|\frac{1-\tan x}{1+\tan x}\right|+c$ (2) $\log\left|\frac{1+\tan x}{1-\tan x}\right|+c$

(3) $\frac{1}{2}\log\left|\frac{\tan x+1}{\tan x-1}\right|+c$ (4) $\frac{1}{2}\log\left|\frac{\tan x-1}{\tan x+1}\right|+c$

(20) $\int e^{-7x}\sin 5x\,dx$ is

(1) $\frac{e^{-7x}}{74}[-7\sin 5x-5\cos 5x]+c$ (2) $\frac{e^{-7x}}{74}[7\sin 5x+5\cos 5x]+c$

(3) $\frac{e^{-7x}}{74}[7\sin 5x-5\cos 5x]+c$ (4) $\frac{e^{-7x}}{74}[-7\sin 5x+5\cos 5x]+c$

(21) $\int x^2 e^{\frac{x}{2}}dx$ is

(1) $x^2e^{\frac{x}{2}}-4xe^{\frac{x}{2}}-8e^{\frac{x}{2}}+c$ (2) $2x^2e^{\frac{x}{2}}-8xe^{\frac{x}{2}}-16e^{\frac{x}{2}}+c$

(3) $2x^2e^{\frac{x}{2}}-8xe^{\frac{x}{2}}+16e^{\frac{x}{2}}+c$ (4) $x^2\frac{e^{\frac{x}{2}}}{2}-\frac{xe^{\frac{x}{2}}}{4}+\frac{e^{\frac{x}{2}}}{8}+c$

(22) $\int \frac{x+2}{\sqrt{x^2-1}}dx$ is

(1) $\sqrt{x^2-1}-2\log\left|x+\sqrt{x^2-1}\right|+c$ (2) $\sin^{-1}x-2\log\left|x+\sqrt{x^2-1}\right|+c$

(3) $2\log\left|x+\sqrt{x^2-1}\right|-\sin^{-1}x+c$ (4) $\sqrt{x^2-1}+2\log\left|x+\sqrt{x^2-1}\right|+c$

(23) $\int \frac{1}{x\sqrt{(\log x)^2-5}}dx$ is

(1) $\log\left|x+\sqrt{x^2-5}\right|+c$ (2) $\log\left|\log x+\sqrt{\log x-5}\right|+c$

(3) $\log\left|\log x+\sqrt{(\log x)^2-5}\right|+c$ (4) $\log\left|\log x-\sqrt{(\log x)^2-5}\right|+c$

(24) $\int \sin\sqrt{x}dx$ is

(1) $2\left(-\sqrt{x}\cos\sqrt{x}+\sin\sqrt{x}\right)+c$ (2) $2\left(-\sqrt{x}\cos\sqrt{x}-\sin\sqrt{x}\right)+c$

(3) $2\left(-\sqrt{x}\sin\sqrt{x}-\cos\sqrt{x}\right)+c$ (4) $2\left(-\sqrt{x}\sin\sqrt{x}+\cos\sqrt{x}\right)+c$

(25) $\int e^{\sqrt{x}}dx$ is

(1) $2\sqrt{x}(1-e^{\sqrt{x}})+c$ (2) $2\sqrt{x}(e^{\sqrt{x}}-1)+c$

(3) $2e^{\sqrt{x}}(1-\sqrt{x})+c$ (4) $2e^{\sqrt{x}}(\sqrt{x}-1)+c$

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